Практика Логирование и мониторинг

Запуск Prometheus и Grafana:

```
red57@compute-vm-8-16-30-ssd-1772094666092:~/prometheus$ docker run -d \
  --name prometheus_red57 \
  --network demo-network \
  -p 9057:9090 \
  -v /home/red57/prometheus/prometheus.yml:/etc/prometheus/prometheus.yml \
  -v /home/red57/prometheus/alerts.yml:/etc/prometheus/alerts.yml \
  -v prometheus_data_red57:/prometheus \
  prom/prometheus
4b726184a02b8ad2763d1fa47ee58c1edf15baed67b0e697d9096214754e83f0
red57@compute-vm-8-16-30-ssd-1772094666092:~/prometheus$ docker ps -a
CONTAINER ID   IMAGE                          COMMAND                  CREATED         STATUS
                  PORTS                                                                  NAMES
4b726184a02b   prom/prometheus                "/bin/prometheus --c…"   8 seconds ago   Up 7 seconds
                  0.0.0.0:9057->9090/tcp, [::]:9057->9090/tcp                             prometheus_red57
```

```
red57@compute-vm-8-16-30-ssd-1772094666092:~/grafana$ docker-compose up -d
WARN[0000] /home/red57/grafana/docker-compose.yml: the attribute `version`
[+] up 5/5
 ✔ Container loki28     Recreated
 ✔ Container grafana28  Recreated
 ✔ Container promtail57 Started
 ✔ Container loki57     Started
 ✔ Container grafana57  Started
```

Контейнеры запущены:

```
red57@compute-vm-8-16-30-ssd-1772094666092:~$ docker ps -a
CONTAINER ID   IMAGE                          COMMAND                  CREATED
       STATUS                          PORTS
                     NAMES
de385c249984   grafana/promtail:latest        "/usr/bin/promtail -…"   38 minutes ag
o      Up 38 minutes
                     promtail57
929e4d6a1ff7   grafana/loki:latest            "/usr/bin/loki -conf…"   38 minutes ag
o      Up 38 minutes                   0.0.0.0:3157->3100/tcp, [::]:3157->3100/tcp
                     loki57
8e85c60b619e   grafana/grafana:latest         "/run.sh"                38 minutes ag
o      Up 38 minutes                   0.0.0.0:3057->3000/tcp, [::]:3057->3000/tcp
                     grafana57
c91f0b4a1ed7   demo-prometeus-app-app         "java -jar app.jar"      About an hour
 ago   Up About an hour                0.0.0.0:8157->8080/tcp, [::]:8157->8080/tcp
                     app_red57
4b726184a02b   prom/prometheus                "/bin/prometheus --c…"   2 hours ago
       Up 48 minutes                   0.0.0.0:9057->9090/tcp, [::]:9057->9090/tcp
                     prometheus_red57
```

Структура файлов:

```
red57@compute-vm-8-16-30-ssd-1772094666092:~$ tree
.
├── demo-prometeus-app
│   ├── Dockerfile
│   ├── build.gradle
│   ├── docker-compose.yml
│   ├── gradle
│   │   └── wrapper
│   │       ├── gradle-wrapper.jar
│   │       └── gradle-wrapper.properties
│   ├── gradlew
│   ├── gradlew.bat
│   ├── settings.gradle
│   └── src
│       └── main
│           ├── java
│           │   └── com
│           │       └── nexign
│           │           └── demoPrometeusApp
│           │               ├── App.java
│           │               └── controller
│           │                   └── EchoController.java
│           └── resources
│               └── application.yml
├── grafana
│   ├── docker-compose.yml
│   ├── loki-config.yaml
│   └── promtail-config.yaml
└── prometheus
    ├── alerts.yml
    └── prometheus.yml

14 directories, 16 files
```

```
red57@compute-vm-8-16-30-ssd-1772094666092:~/grafana$ cat docker-compose.yml
version: '3'

services:
  grafana:
    image: grafana/grafana:latest
    container_name: grafana57
    ports:
      - "3057:3000"
    environment:
      - GF_SECURITY_ADMIN_PASSWORD=admin
      - GF_FEATURE_TOGGLES_ENABLE=loki
    volumes:
      - grafana_data:/var/lib/grafana
    networks:
      - demo-network
    restart: unless-stopped

  loki:
    image: grafana/loki:latest
    container_name: loki57
    ports:
      - "3157:3100"
    command: -config.file=/etc/loki/local-config.yaml
    volumes:
      - ./loki-config.yaml:/etc/loki/local-config.yaml
      - loki_data:/loki
    networks:
      - demo-network
    restart: unless-stopped

  promtail:
    image: grafana/promtail:latest
    container_name: promtail57
    volumes:
      - ./promtail-config.yaml:/etc/promtail/config.yml
      - /var/run/docker.sock:/var/run/docker.sock
      - /var/lib/docker/containers:/var/lib/docker/containers:ro
      - /var/log:/var/log:ro
    command: -config.file=/etc/promtail/config.yml
    networks:
      - demo-network
    restart: unless-stopped
    depends_on:
      - loki

volumes:
  grafana_data:
  loki_data:

networks:
  demo-network:
    external: true
```

```
red57@compute-vm-8-16-30-ssd-1772094666092:~/grafana$ cat loki-config.yaml
auth_enabled: false

server:
  http_listen_port: 3100

common:
  path_prefix: /loki
  storage:
    filesystem:
      chunks_directory: /loki/chunks
      rules_directory: /loki/rules
  replication_factor: 1
  ring:
    kvstore:
      store: inmemory

schema_config:
  configs:
    - from: 2020-10-24
      store: tsdb
      object_store: filesystem
      schema: v13
      index:
        prefix: index_
        period: 24h
red57@compute-vm-8-16-30-ssd-1772094666092:~/grafana$ cat promtail-config.yaml
server:
  http_listen_port: 9080
  grpc_listen_port: 0

positions:
  filename: /tmp/positions.yaml

clients:
  - url: http://loki57:3100/loki/api/v1/push

scrape_configs:
  - job_name: containers
    docker_sd_configs:
      - host: unix:///var/run/docker.sock
        refresh_interval: 15s
    relabel_configs:
      - source_labels: ['_meta_docker_container_name']
        target_label: 'container'
      - source_labels: ['_meta_docker_container_log_stream']
        target_label: 'stream'
      - source_labels: ['_meta_docker_container_id']
        target_label: 'docker_id'
```

```
red57@compute-vm-8-16-30-ssd-1772094666092:~/prometheus$ cat prometheus.yml
global:
  scrape_interval: 15s

rule_files:
  - "alerts.yml"

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

  - job_name: 'node'
    static_configs:
      - targets: ['178.154.195.14:9100']

  - job_name: 'app'
    metrics_path: '/actuator/prometheus'
    static_configs:
      - targets: ['178.154.195.14:8157']
red57@compute-vm-8-16-30-ssd-1772094666092:~/prometheus$ cat alerts.yml
groups:
  - name: cpu_alerts
    rules:
      - record: instance:cpu_usage:rate5m
        expr: rate(node_cpu_seconds_total{mode!="idle"}[5m])

      - alert: HighCPUUsage
        expr: instance:cpu_usage:rate5m > 0.8
        for: 1m
        labels:
          severity: warning
        annotations:
          summary: "Высокая загрузка CPU {{ $labels.instance }}"
red57@compute-vm-8-16-30-ssd-1772094666092:~/prometheus$
```

Интерфейс Prometheus

Query | Alerts | Status > Target health

Select scrape pool | Filter by target health | Filter by endpoint or labels

app — 1 / 1 up

Endpoint	Labels	Last scrape	State
http://178.154.195.14:8157/actuator/prometheus	instance="178.154.195.14:8157" job="app"	10.095s ago 6ms	UP

node — 1 / 1 up

Endpoint	Labels	Last scrape	State
http://178.154.195.14:9100/metrics	instance="178.154.195.14:9100" job="node"	12.501s ago 77ms	UP

prometheus — 1 / 1 up

Endpoint	Labels	Last scrape	State
http://localhost:9090/metrics	instance="localhost:9090" job="prometheus"	10.461s ago 4ms	UP

Query | Alerts | Status

Filter by rule group state | Filter by rule name or labels

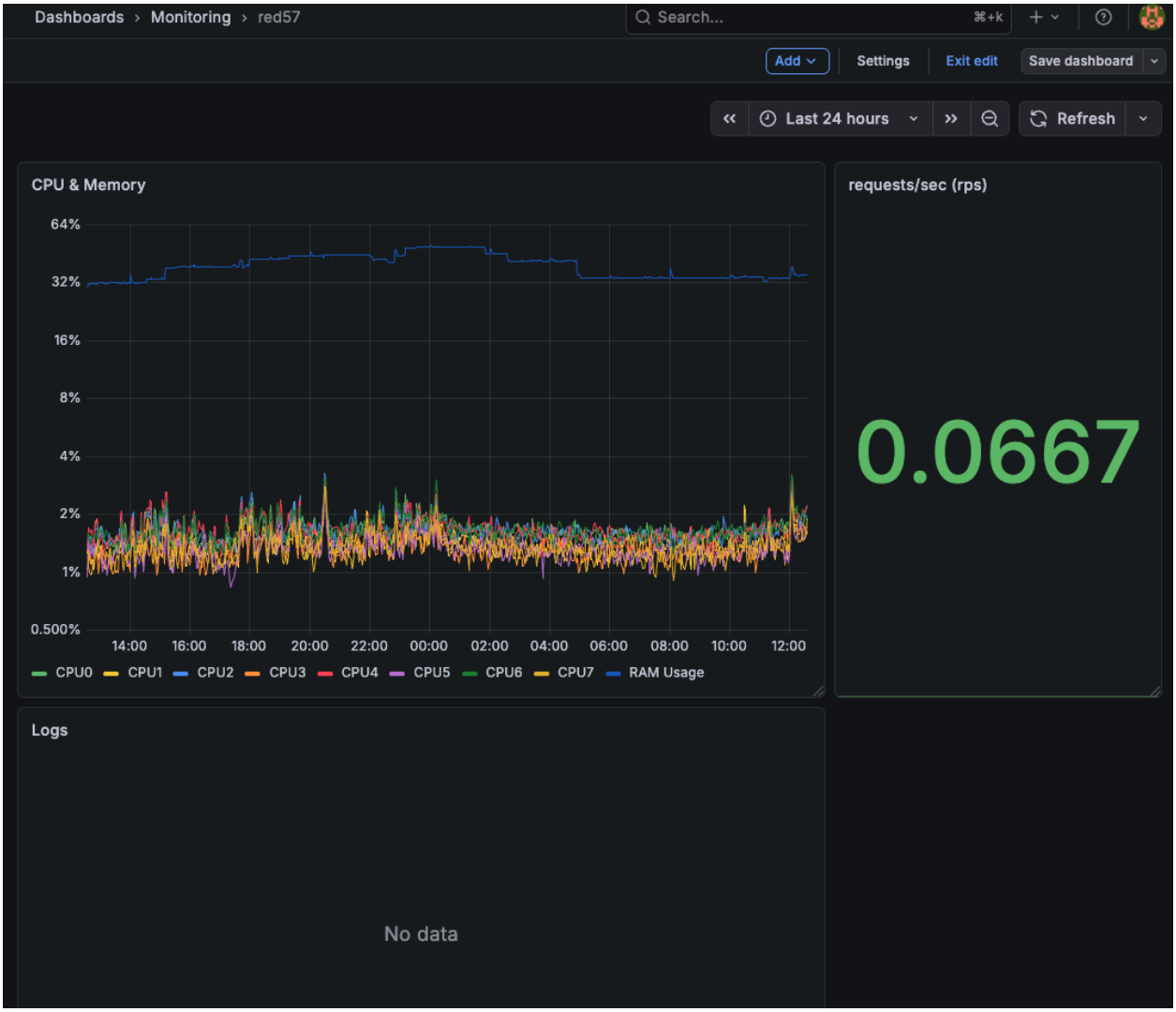
cpu_alerts /etc/prometheus/alerts.yml — INACTIVE (1)

HighCPUUsage

```
instance:cpu_usage:rate5m > 0.8
```

for: 1m

severity="warning"

summary: Высокая загрузка CPU {{ $labels.instance }}

Дашборд в Grafana